

# Congenital & Extrapulmonary Tuberculosis

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# Outlines



Congenital TB



PTB VS EPTB



TB of Lymph nodes



Pleural Tuberculosis



TB Meningitis



Skeletal Tuberculosis



Treatment



# Congenital Tuberculosis

## \*Congenital Tuberculosis \*

- Refers to the acquisition of Mycobacterium tuberculosis either **in utero** or during the **intrapartum** period
- High neonatal mortality (up to 60%) and morbidity warrant early diagnosis and treatment
- True congenital tuberculosis: route of transmission
  1. **Transplacental through umbilical vein** from mother with primary hematogenous tuberculosis during pregnancy. Liver is enlarged, enlarge lymph node at porta hepatis, and evidence of miliary disease may seen in infant
  2. **In utero aspiration of amniotic fluid** infected from endometritis in the mother or from placenta

# \*Congenital Tuberculosis \*

Other route of transmission: Perinatal Tuberculosis (Congenital and Postnatal):

- Ingestion of infected amniotic fluid or secretion during delivery
- Inhalation of tubercle bacilli at or soon after birth from mother, other relatives, or attendants with infection pulmonary tuberculosis. (most common mode of transmission in neonate)



# ✿ Clinical Presentations ✿

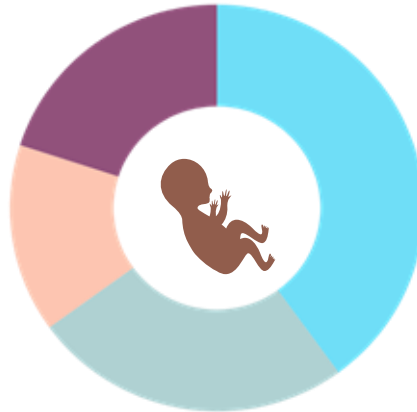
Difficulty in recognizing as symptoms and signs may be minimal or even absent in the newborn  
Symptoms usually **develop 2–3 weeks after birth**

## ✿ Very common(>60%)

- Respiratory distress
- Hepatomegaly
- Fever (usually low grade)

## Common (40-60%)

- Cough
- Prematurity/growth failure
- Splenomegaly



## Frequent (25-40%)

- Poor feeding
- Abdominal distension (including ascites)

## Infrequent (10-25%)

- Irritability and lethargy
- Peripheral lymphadenopathy
- Sepsis syndrome
- Seizure

**Rare (<10%):** Skin papular/pustular or ulcerative lesions, jaundice (obstructive), otorrhea/mastoiditis

# Congenital Tuberculosis



| Type of TB (can co-exist)                                                              | Occurrence |
|----------------------------------------------------------------------------------------|------------|
| ▪ Pulmonary TB including intrathoracic lymph nodes                                     | >90%       |
| ▪ Miliary TB                                                                           | 25 – 35%   |
| ▪ Abdominal TB (hepatic, splenic, peroneal lesions and ascites, intra-abdominal nodes) | 20 – 40%   |
| ▪ Peripheral lymph node TB                                                             | 10 – 20%   |
| ▪ TB meningitis                                                                        | 10 – 20%   |
| ▪ Ear/mastoid TB                                                                       | 5 – <10%   |
| ▪ Skin TB                                                                              | 5 – <10%   |



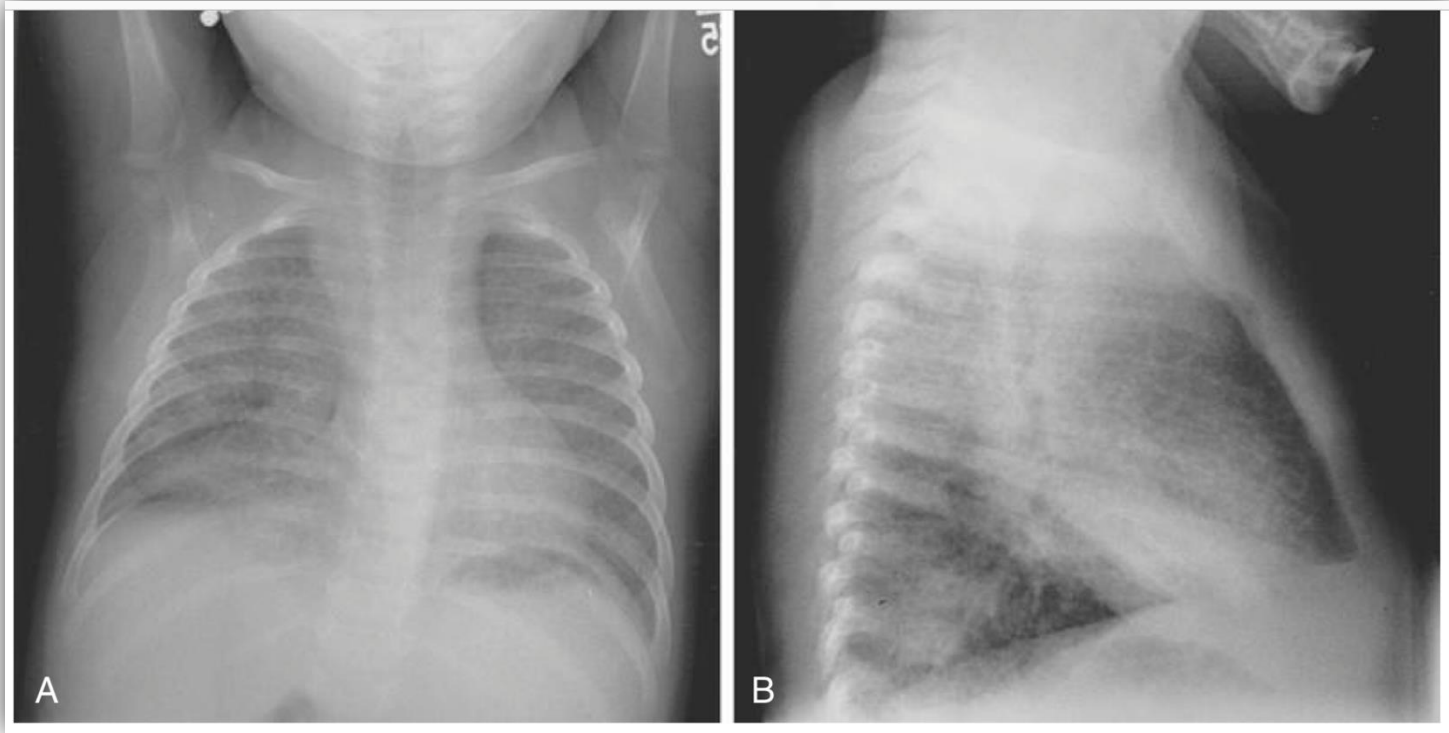
# Chest X-ray Features

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|                                          |        |
|------------------------------------------|--------|
| ▪ Alveolar opacification                 | 30–40% |
| ▪ Miliary infiltrates                    | 30–40% |
| ▪ Bilateral bronchopneumonic infiltrates | 15–20% |
| ▪ Intrathoracic lymph nodes              | 10–15% |
| ▪ Pleural effusion                       | <5%    |
| ▪ Cavities                               | <5%    |
| ▪ Normal chest x-ray                     | 5–10%  |

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Milliary tuberculosis in an infant

# Cantwell Revised Criteria

A **confirmed TB lesion** and **one or more** of the following

- Lesions presents within the 1<sup>st</sup> week of life
- A **primary hepatic complex or caseating hepatic granuloma**
- Mycobacterial bacilli identified in the **placenta or endometrial TB** in the mother
- **Excluding possible postnatal transmission** by thorough investigation of contacts

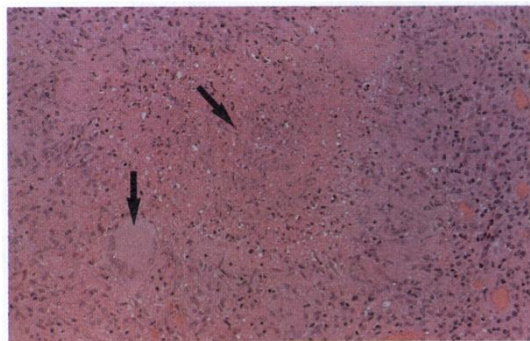


Figure 1. Endometrial-Biopsy Specimen from the Mother of Patient 1, Showing a Caseating Granuloma (Center Arrow) and a Langhans' Giant Cell (Lower-Left Arrow) (Hematoxylin and Eosin,  $\times 20$ ).

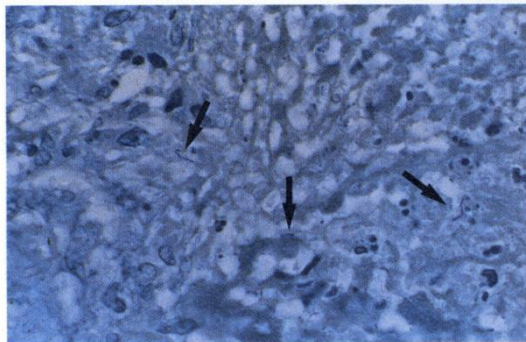
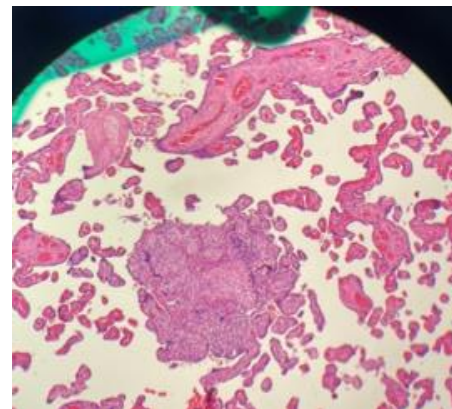


Figure 2. Placenta of Patient 1, Showing Acid-Fast Bacilli (Arrows) and Inflammatory Cells in an Intervillous Thrombus (Ziehl-Neelsen,  $\times 100$ ).



# Management of Congenital TB

- Treatment should be started based on the likelihood of TB, **even before bacteriological confirmation is received**, as **TB can progress rapidly in neonate**
- If PTB or tuberculous peripheral lymphadenitis is suspected/confirmed in these infants, they should be treated promptly with a **6 - month treatment regimen (2HRZ(E)/4HR)**

# Management of Congenital TB

- Favorable response to treatment: increased appetite, weight gain and radiographic resolution
- **Breastfeeding is recommended**, irrespective of the mother's TB status
  - Risk of TB transmission through breast milk is negligible
  - Commonly used TB medicines are excreted into breast milk in small amounts, there is no evidence that this induces drug resistance
- **Separation from the mother is not advised**, especially in resource-limited settings, where establishing breastfeeding can be critical for the child's survival

# Management of Asymptomatic Neonates of Mothers with TB

- TB disease should be excluded in neonates born to women with presumptive or confirmed TB
- It is not necessary to separate the neonate from the mother
- Breastfeeding should be continued (mother advised to wear a surgical mask when close to the baby)
- While screening for TB disease or TB infection is ongoing, **BCG should be postponed in neonates exposed to TB**
  - BCG will interfere with the interpretation of TST, reducing the effectiveness of the test for diagnosing infection

# Management of Asymptomatic Neonates of Mothers with TB

- If the newborn is well and without any signs or symptoms of TB, provide TPT **(preferably 3HR)**
- Administer pyridoxine 5–10 mg/day
- If the infant is exposed to HIV (e.g. mother living with HIV) and is on nevirapine
  - Isoniazid preventive treatment should be started
  - TPT with rifamycins (e.g. 3HR or 3HP) should not be given with nevirapine prophylaxis

# Tuberculosis Preventive Treatment (TPT) Regimens



| Recommendation                                                                                                                                                                                                                                                                           | Alternative                                                                                                                                                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• 6 months or 9 months of isoniazid daily (6H or 9H) (all ages)</li><li>• 3 months of isoniazid plus rifapentine weekly (3HP) (age <math>\geq</math> 2 years old)</li><li>• 3 months of isoniazid plus rifampicin daily (3HR) (all ages)</li></ul> | <ul style="list-style-type: none"><li>• 1 month of daily isoniazid plus rifapentine (1HP) (age <math>\geq</math> 13 years old)</li><li>• 4 months of daily rifampicin (4R) (all ages)</li></ul> |



## Key point: preferred TB preventive treatment options

3HR is a preferred TPT option among HIV-negative children since child-friendly dispersible FDCs are widely available and already used for TB treatment, while awaiting data on dosages across all age groups and child-friendly formulations for rifapentine-based regimens. For children living with HIV, 6H remains the preferred option until further data are available.

# Tuberculosis Preventive Treatment (TPT) Regimens

Non-HIV

HIV

| กลุ่ม                                     | สูตรแนะนำ                                      | สูตรทางเลือก                            |
|-------------------------------------------|------------------------------------------------|-----------------------------------------|
| อายุ < 2 ปี<br>ที่ไม่ติดเชื้อเอชไอวี      | 3HR หรือ 6-9H                                  | 4R                                      |
| อายุ ≥ 2 ปี<br>ที่ไม่ติดเชื้อเอชไอวี      | 3HR หรือ 3HP หรือ<br>6-9H                      | 4R                                      |
| เด็ก < 13 ปี<br>ที่ติดเชื้อเอชไอวี        | 6-9H                                           | 3HR หรือ 3HP ถ้า<br>รับประทานยาสูตร EFV |
| วัยรุ่นอายุ ≥ 13 ปี<br>ที่ติดเชื้อเอชไอวี | 3HP ถ้ากินยาสูตร TDF,<br>EFV, DTG, RAL<br>6-9H | 1HP ถ้ากินยาสูตร TDF,<br>EFV, DTG, RAL  |



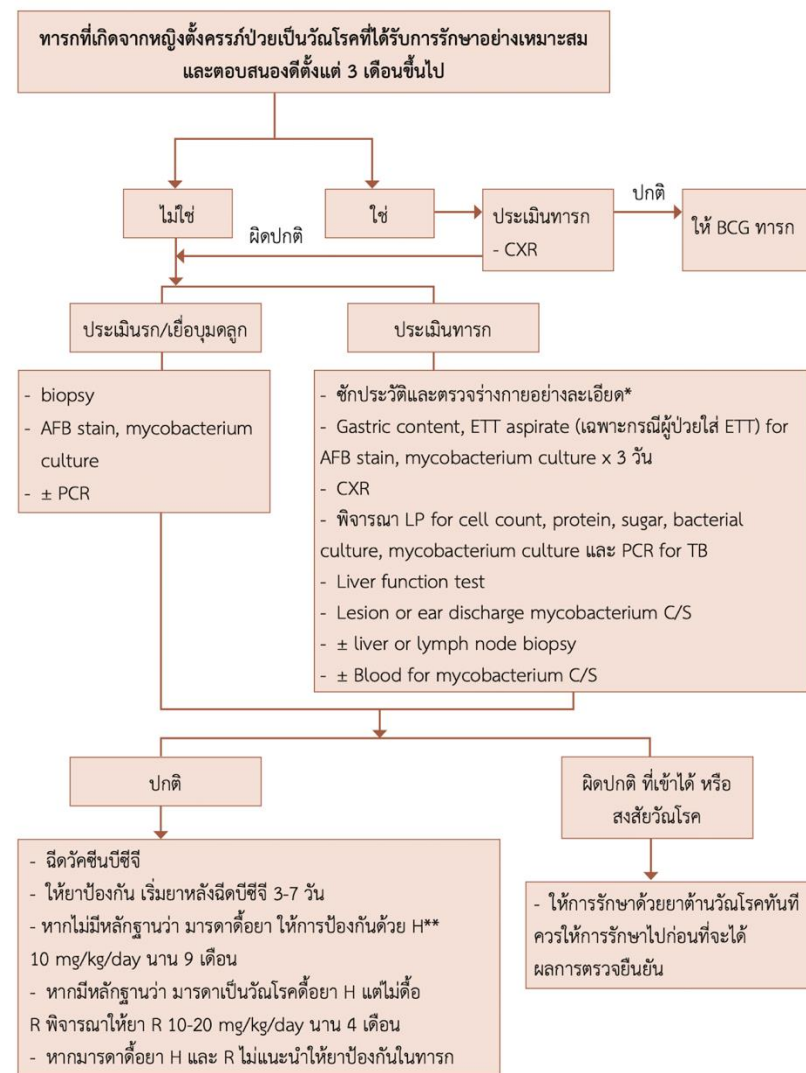
# Management of Asymptomatic Neonates of Mothers with TB

- Regularly followed up and monitored for the development of symptoms and signs suggestive of TB
- If the infant remains asymptomatic after completion of TPT, **TST or IGRA** should be performed if available
  - Negative or not available: give BCG (unless the baby is living with HIV) **2 weeks after completion of the full course of TPT**
  - Positive: BCG is not required

# Thai National Tuberculosis Control Program Guildline

## การแยกมารดาและการให้นมบุตร

เนื่องจากการติดเชื้อหลังคลอดนี้ไม่ได้แพร่โดยการกินนมมารดา เชื้อวัณโรคไม่พบในน้ำนม แม้ว่ายาต้านวัณโรคที่มารดากินอาจออกมาทางน้ำนมได้บ้างแต่ก็มีในปริมาณน้อยจนไม่เกิดอันตรายต่อทารก และไม่มีผลต่อการรักษาในทารก ดังนั้นทารกสามารถกินนมมารดาที่เป็นวัณโรคได้ อย่างไรก็ตามมารดาต้องอยู่ในระยะที่แพร่เชื้อได้ คือในช่วง 2 สัปดาห์แรกของการรักษาหรือเสมหะยังเป็นบวก ควรให้กินนมมารดาที่ปั๊พออกมา (expressed milk) มากกว่าการให้กินจากเต้า และควรให้ทารกแยกจากมารดาในช่วงที่มีโอกาสแพร่เชื้อ



| Table II. Comparison of recommendations on isolation, breast-feeding and vaccination by different groups |                                                               |                                     |                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name of the group                                                                                        | Breast feeding                                                | Barrier method                      | Isolation                                                                                                                                                                 | BCG Vaccination                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                |
| IAP <sup>18</sup>                                                                                        | Can continue to breast feed.                                  | Cough hygiene                       | Not required if mother on treatment. Isolation if mother hospitalized, non-adherent to therapy or has MDR-TB.                                                             | Advised- at birth even with INH prophylaxis (IAP 2012)                                                                                                                                                                                                                                                                                                                                                                                                         | Not mentioned.                                                                                                                                                                                                                 |
| DOTS <sup>19</sup>                                                                                       | Feeding encouraged if mother sputum negative                  | Face mask                           | If mother has active disease and is noncompliant to treatment or has received ATT prior to delivery.                                                                      | Vaccination may be postponed or done with INH resistant BCG vaccine.                                                                                                                                                                                                                                                                                                                                                                                           | Not mentioned.                                                                                                                                                                                                                 |
| AAP <sup>20</sup>                                                                                        | Breast feeding if mother is on ATT.                           | Face mask                           | Separation avoided. Advised if MDR-TB, non-compliant to therapy, mother has contagious TB before starting ATT.                                                            | Give BCG if mother has MDRTB, or poorly adherent to treatment                                                                                                                                                                                                                                                                                                                                                                                                  | Not mentioned.                                                                                                                                                                                                                 |
| WHO <sup>21</sup>                                                                                        | To continue                                                   | Face mask                           | If mother has MDR-TB.                                                                                                                                                     | Delay BCG until INH therapy is completed. BCG after 2 weeks of completing therapy                                                                                                                                                                                                                                                                                                                                                                              | Question not answered under level evidence                                                                                                                                                                                     |
| New Zealand Guidelines On Tuberculosis <sup>22</sup>                                                     | Recommended irrespective of the maternal status               | Mask and infection control measures | Separation till mother fully evaluated or until both are on treatment, if mother has MDR-TB or poor adherence to therapy. infant on isoniazid therapy needs no separation | BCG after 3 months of INH therapy                                                                                                                                                                                                                                                                                                                                                                                                                              | Not mentioned.                                                                                                                                                                                                                 |
| MERK MANUAL <sup>23</sup>                                                                                | Breast feeding not contraindicated                            | =                                   | Mother and baby should stay together.                                                                                                                                     | If mother non adherent BCG should be given.                                                                                                                                                                                                                                                                                                                                                                                                                    | Not mentioned.                                                                                                                                                                                                                 |
| MTS <sup>24</sup>                                                                                        | Give expressed breast milk till mother sputum smear negative. | -                                   | If mother smear positive at time of delivery.                                                                                                                             | BCG at birth if congenital TB ruled out. (level III) BCG should not be given to babies on prophylactic TB treatment(Grade C) After completing INH prophylaxis for 3 months mountoux if negative BCG administered or if positive BCG at 6 months. In babies born to mothers with active PTB < 2 months after deliver BCG may be repeated but no role of reimmunizing babies if mothers diagnosed of active pulmonary TB after 2 months of delivery. (level III) | Grade C recommendation defined as evidence from expert committee Reports , opinion and/or clinical experience of respected authorities, directly Indicates the absence of directly applicable clinical studies of good quality |

Contd....

| Name of the group    | Breast feeding                             | Barrier method | Isolation                               | BCG Vaccination                                                 |                |
|----------------------|--------------------------------------------|----------------|-----------------------------------------|-----------------------------------------------------------------|----------------|
| SASPID <sup>25</sup> | Mother should be encouraged to breast feed | -              | -                                       | BCG after 6 months of INH therapy or after completing treatment | Not mentioned. |
| NICE <sup>26</sup>   | Breast feeding allowed                     | -              | Mother and baby should not be separated | -                                                               | Not mentioned. |

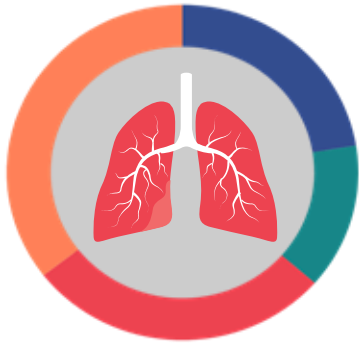
AAP, American Academy of Pediatrics; ATT, Antitubercular Treatment; BCG, Bacille Calmette-Guérin Vaccine; CDC, Centers for Disease Control and Prevention; DOTS, Directly Observed Treatment - short course; MDR-TB, Multi Drug Resistant Tuberculosis; MTx, Mantoux Test; PPD, Purified Protein Derivative Test (Tuberculin test); WHO, World Health Organization; MTS, Malaysian Thoracic Society; NICE, National Institute for Health and Clinical Excellence



# Pulmonary VS Extrapulmonary Tuberculosis

# Definition of Pulmonary TB

PTB: any bacteriologically confirmed or clinically diagnosed case of TB involving the **lung parenchyma** or **tracheobronchial tree**



**Tuberculous mediastinal and/or hilar intrathoracic lymphadenopathy** classified as **PTB**

**Miliary TB** is classified as PTB because there are lesions in the lungs

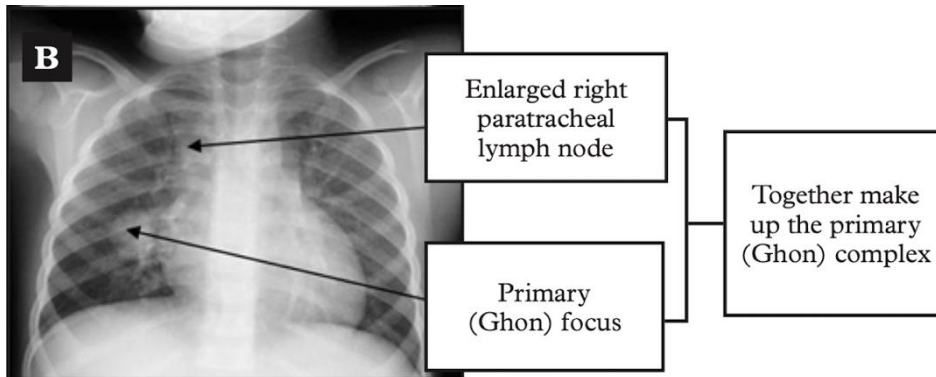
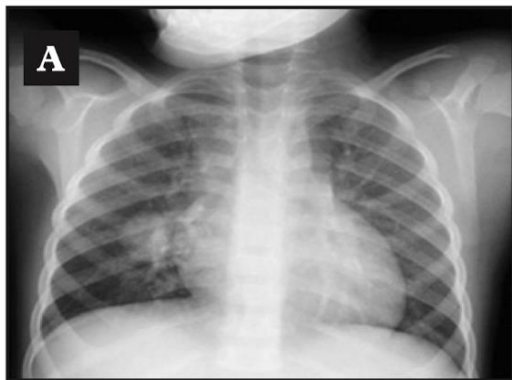
Tuberculous pleural effusion without radiographic abnormalities in the lungs constitutes EPTB

A person with both PTB and EPTB should be classified as having PTB

# Abnormalities on CXR suggestive of Pulmonary TB

## Primary (Ghon) focus and complex

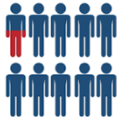
- Primary focus rarely seen on CXR
- Enlarged mediastinal LN commonly seen on CXR
- Primary focus + Enlarged mediastinal LN → primary complex or Ghon complex uncommon



# Abnormalities on CXR suggestive of Pulmonary TB

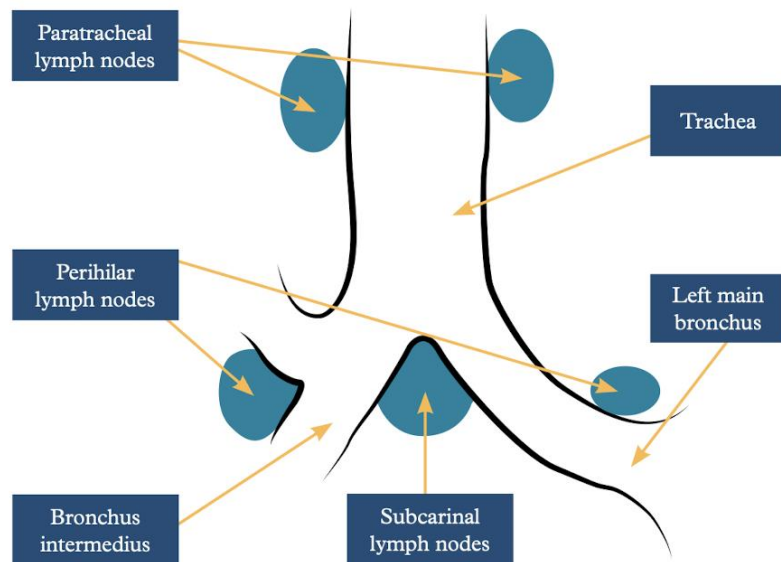
## The primary (Ghon) focus

- Uncommon CXR feature
- Can occur in any of the lobes, tends to occur **1-2 cm from the pleura**
- Normally poorly circumscribed, less than 1 cm in diameter
- Can have more than one primary focus and an overlying pleural reaction may be visible on CXR

|                                                                                                                      |                                                                                     |               |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------|
| <b>The primary (Ghon) focus</b><br> |  | Very uncommon |
|                                                                                                                      |  | Very specific |
|                                                                                                                      | <b>NON-SEVERE</b>                                                                   | Non-severe    |

# Abnormalities on CXR suggestive of Pulmonary TB

## Mediastinal LN enlargement



Anatomical position of mediastinal lymph nodes

- **Hallmark feature** of pediatric PTB
- Paratracheal and hilar (includes sub-carinal) lymph nodes: mediastinal lymph nodes
- More common in young children (<5 years), less common with increasing age

If there are enlarged paratracheal LN in the absence of associated perihilar or sub-carinal LN enlargement  
\*\*Differential diagnosis of **lymphoma**

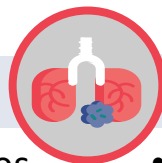
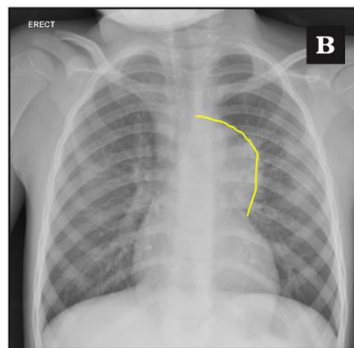
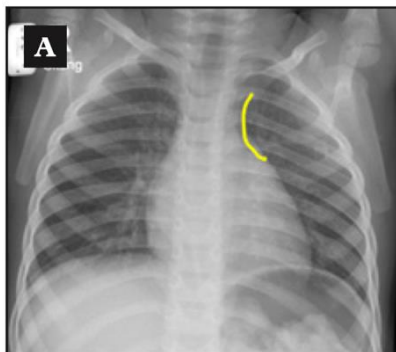


# Abnormalities on CXR suggestive of Pulmonary TB

## Mediastinal LN enlargement

### Perihilar lymph nodes

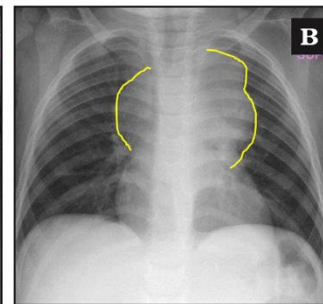
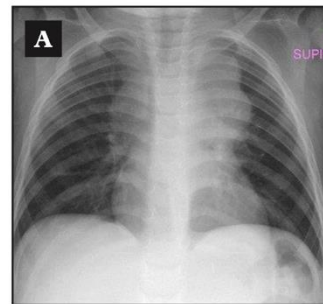
- Lobulated radio-opaque structures in the hilar areas
- Convex shape of the hilum
- More easily seen on lateral film
- Include sub-carinal LN.



### Paratracheal lymph nodes

- Widening, bulging outwards of upper mediastinum

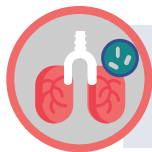
**\*\*complicated in the presence of the thymus in young children (especially <2 years)**



Perihilar & paratracheal lymphadenopathy

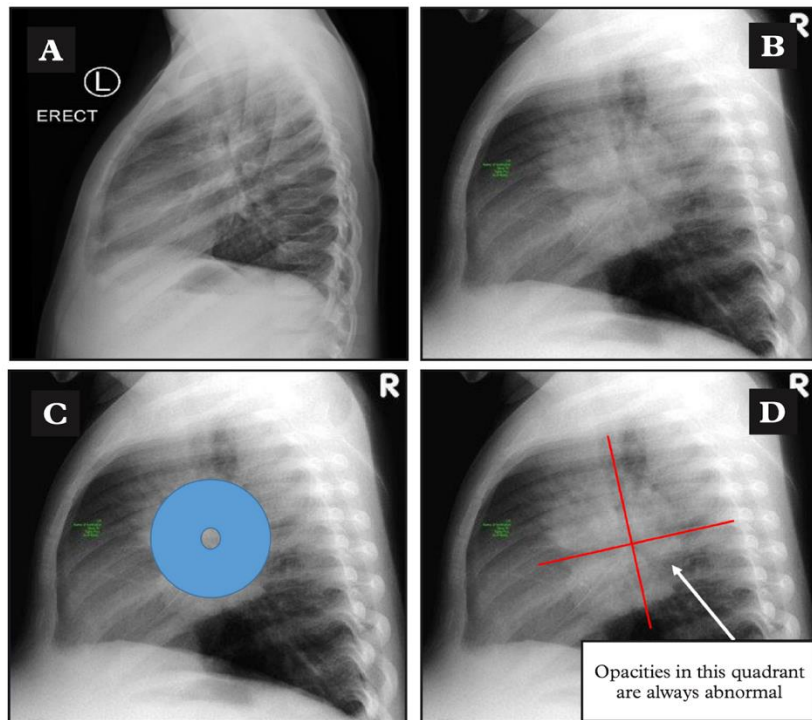
# Abnormalities on CXR suggestive of Pulmonary TB

## Mediastinal LN enlargement



### Using the lateral CXR to identify enlarged mediastinal lymph nodes

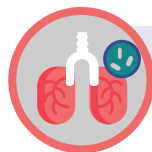
- Sometimes seen on lateral film when they are not clearly seen on the AP/PA film
- Rounded opacities at lower end of trachea **“doughnut” or “hamburger” sign (C)**
- Fill the middle mediastinum Vs thymus fills the anterior mediastinum



## Perihilar lymphadenopathy

# Abnormalities on CXR suggestive of Pulmonary TB

## Mediastinal LN enlargement

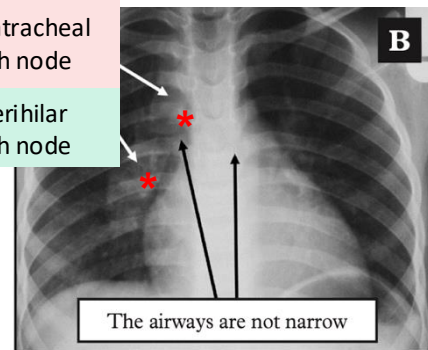


### Uncomplicated lymph node disease

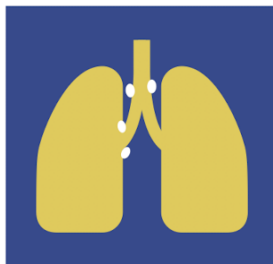
- No significant airway compression AND
- Either minimal (<1 lobe) or no parenchymal involvement

Rt paratracheal lymph node

Rt perihilar lymph node



### Uncomplicated lymph node disease



Very common



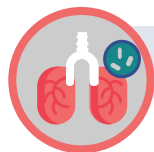
Very specific

NON-SEVERE

Non-severe

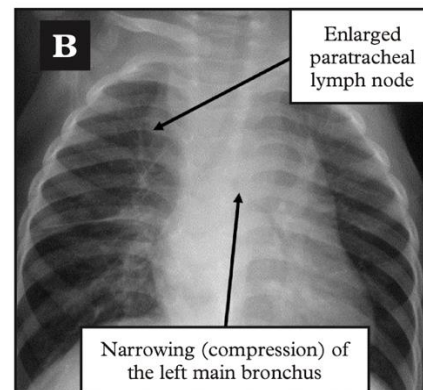
# Abnormalities on CXR suggestive of Pulmonary TB

## Mediastinal LN enlargement



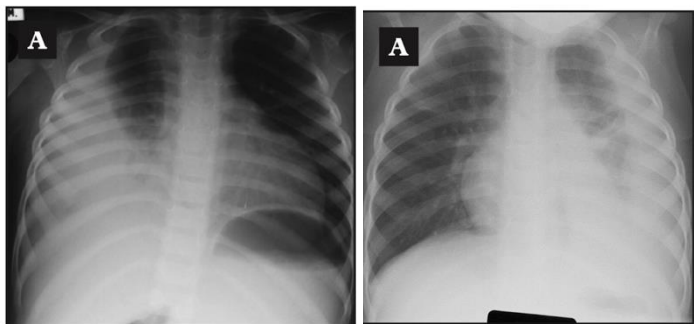
### Complicated lymph node disease

- Airway compression or deviation AND/OR
- Associated hyperinflation/collapse AND/OR
- Extensive ( $\geq 1$  lobe) parenchymal involvement



|                                              |  |               |
|----------------------------------------------|--|---------------|
| <p><b>Complicated lymph node disease</b></p> |  | Uncommon      |
|                                              |  | Very specific |
|                                              |  | Severe        |

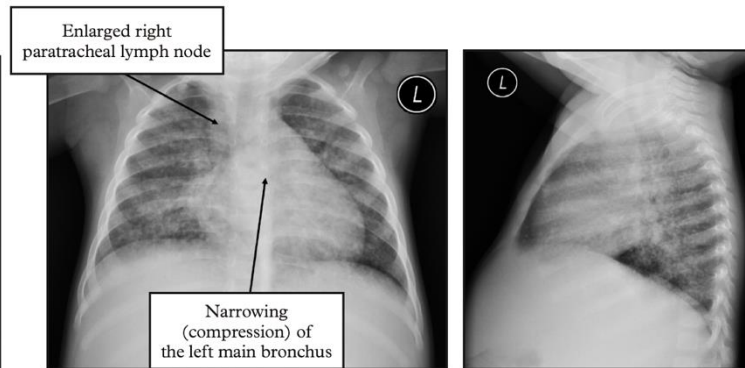
# Abnormalities on CXR suggestive of Pulmonary TB



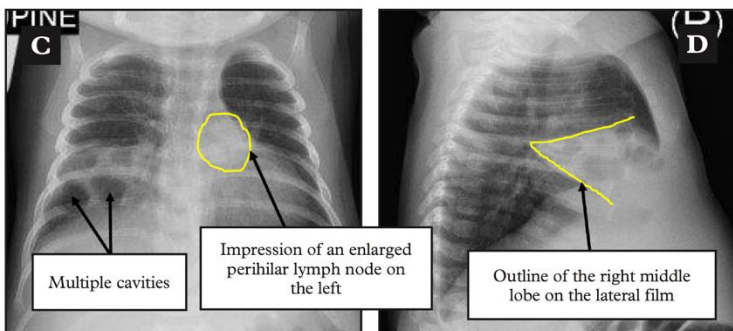
Pleural effusion



Miliary infiltration



Miliary infiltration + paratracheal lymphadenopathy



Cavity lesion

## Key points: CXR findings

- CXR findings in children with PTB are often nonspecific. A CXR alone cannot be used to determine the correct treatment for the child.
- CXR is useful to support the clinical diagnosis of PTB when TB is suspected and bacteriological testing is negative.

# Diagnosis of Pulmonary TB



|                       |                                                                                                                                                                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A. 1+2+3 criteria     | AFB and culture for mycobacterium x 3 specimens, Xpert MTB/RIF 1 specimen and start Rx TB                                                                                                                                                                   |
| B. 1+2 but normal CXR | Evaluate extrapulmonary TB / consult expert                                                                                                                                                                                                                 |
| C. 1+3 criteria       | Treat as bacterial pneumonia +/- atypical pneumonia<br>F/U 1-2 weeks if not improve -> Rx TB, except have typical CXR<br>e.g.miliary infiltration can start Rx pulmonary TB, but need to F/U<br>diagnostic investigation and clinical response to treatment |

# Definition of Extrapulmonary TB

EPTB refers to any bacteriologically confirmed or clinically diagnosed case of TB involving organs other than the lungs (e.g. pleura, **peripheral lymph nodes**, abdomen, genitourinary tract, skin, joints and bones, meninges)

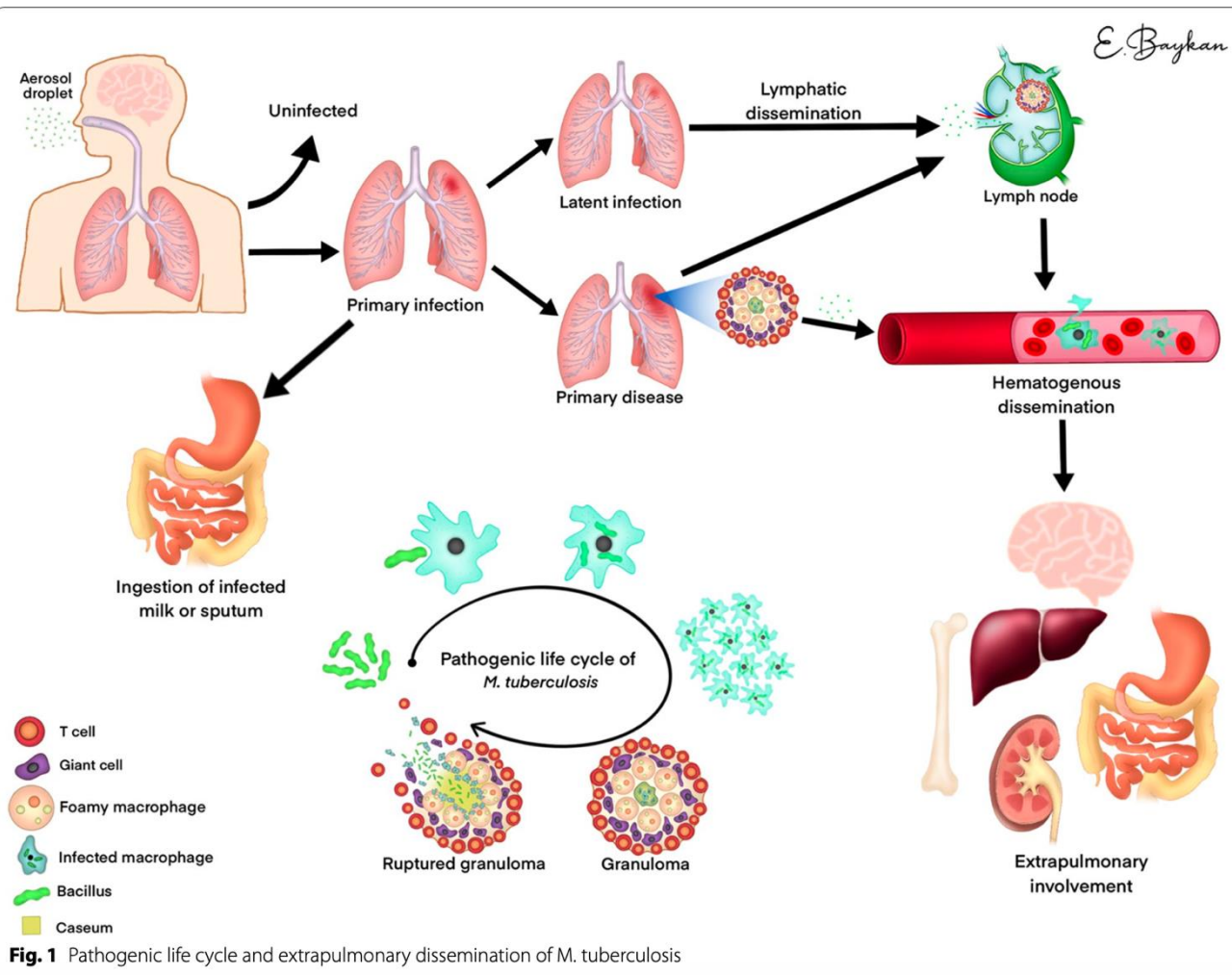


Classification of **intrathoracic lymphadenopathy** in children was updated following an expert consultation in 2021 as **PTB**

**EPTB** is common in young children and in **children and adolescents living with HIV**.

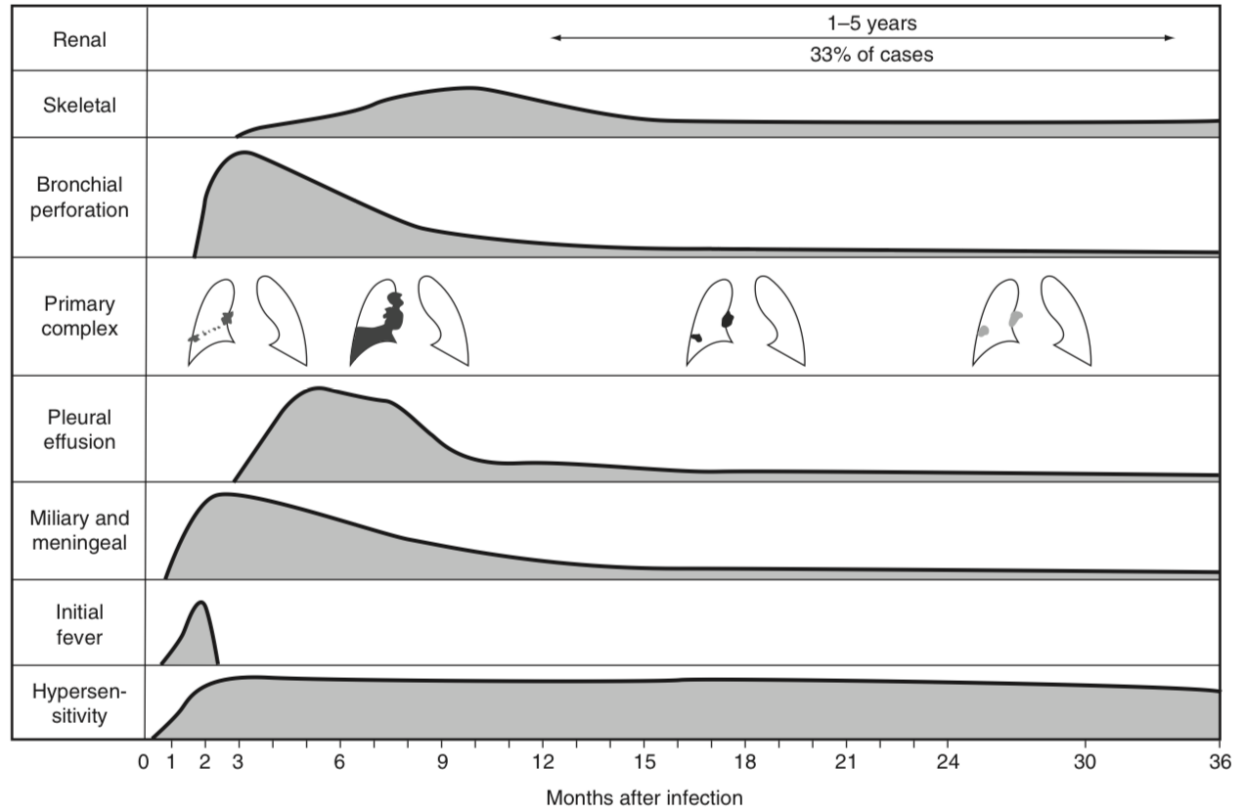
- **Primary** (at the site of initial infection)
- **Secondary** (disseminated): hematogenous or lymphatic spread of bacteria from the primary organ, reactivation of latent TB (LTBI), ingestion of infected sputum, or spread locally from adjacent organs







# Wallgren Timetable of Tuberculosis

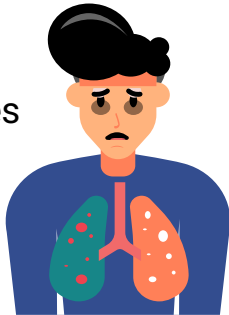


# Median Age of Children with Tuberculosis by Predominant Site of Involvement

| Site          | No. of Cases (%)    | Median Age (y) |
|---------------|---------------------|----------------|
| Pulmonary     | 1213 (77.5)         | 6              |
| Lymphatic     | 209 (13.3)          | 5              |
| Pleural       | 49 (3.1)            | 16             |
| Meningeal     | 29 (1.9)            | 2              |
| Bone or joint | 19 (1.2)            | 8              |
| Other         | 15 (1.0)            | 12             |
| Miliary       | 14 (0.9)            | 1              |
| Genitourinary | 13 (0.8)            | 16             |
| Peritoneal    | 4 (0.3)             | 13             |
| Not stated    | 1 (0.1)             | —              |
| <i>Total</i>  | <i>1566 (100.0)</i> | <i>6</i>       |

# Symptoms of TB in Children

- **Prolong fever**  
with or without night sweats
- **Cough**  
especially if persistent  
and not resolving
- **Anorexia**  
Weight loss 5% in last 3  
months or FTT
- **Unusual fatigue**  
decreased activities



Symptoms that **highly suggestive of extrapulmonary TB**

- Gibbus (especially recent onset) from vertebral TB
- Enlarged and painless joints
- Cervical lymphadenopathy with fistula formation
- Meningitis or pleural effusion that does not respond to antibiotics
- Pericardial effusion, distended abdomen with ascites
- Signs of tuberculin hypersensitivity (e.g., phlyctenular conjunctivitis or erythema nodosum)

**TB lymphadenitis** is common in children. Lymph nodes, usually supraclavicular or cervical, are swollen, painless, and firm

# Typical Clinical Features of Extrapulmonary TB

| Site of EPTB                              | Typical clinical presentation                                                                                                                                                                                                            | Investigations                                                                                                                                                                                                             |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Common forms of EPTB</b>               |                                                                                                                                                                                                                                          |                                                                                                                                                                                                                            |
| TB of lymph nodes<br>(TB lymphadenopathy) | Asymmetrical, painless, non-tender lymph node enlargement (often $>2 \times 2$ cm) for $>1$ month not responding to other treatment (e.g. antibiotics)<br>With or without discharging sinus<br>Most commonly cervical nodes <sup>a</sup> | CXR<br>Ultrasound (if available)<br>Fine-needle aspiration biopsy or excision biopsy with histology, Xpert MTB/RIF or Ultra, culture (if available)<br>Respiratory specimens for Xpert MTB/RIF or Ultra                    |
| Pleural TB                                | Dullness on percussion and reduced breath sounds with or without pleuritic chest pain<br>No acute illness                                                                                                                                | CXR<br>Ultrasound (if available)<br>TST/IGRA<br>Pleural tap with biochemical analysis and cell counts, Xpert MTB/RIF or Ultra or culture (if available) <sup>b,c</sup><br>Respiratory specimens for Xpert MTB/RIF or Ultra |

If axillary node enlargement on same side as BCG vaccination, consider BCG disease

# Typical Clinical Features of Extrapulmonary TB

Usually children aged <5 years with disseminated disease and severely ill

|     |                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                           |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TBM | Subacute onset (>5 days) of headache, irritability or abnormal behaviour, vomiting (without diarrhoea), lethargy, reduced level of consciousness, convulsions, neck stiffness, bulging fontanelle, cranial nerve palsies | CXR<br>Lumbar puncture with biochemical analysis and cell counts and Xpert MTB/RIF or Ultra (rather than culture initially) completed on CSF <sup>b</sup><br>Brain imaging (if available)<br>Respiratory specimens for Xpert MTB/RIF or Ultra<br>TST/IGRA |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                         |                                                                  |                                                                                                                                                                                                                          |
|-------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Miliary TB <sup>d</sup> | Nonspecific symptoms with persistent fever, lethargy and wasting | CXR (typical miliary pattern)<br>Respiratory specimens for Xpert MTB/RIF or Ultra<br>Lumbar puncture with biochemical analysis and cell counts and Xpert MTB/RIF or Ultra on CSF to exclude TBM <sup>b</sup><br>TST/IGRA |
|-------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Miliary TB** is classified as PTB but often involves extrapulmonary features

# Typical Clinical Features of Extrapulmonary TB

| Site of EPTB                         | Typical clinical presentation                                                                                                                                 | Investigations                                                                                                                                                                                               |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Usually children aged $\geq 5$ years |                                                                                                                                                               |                                                                                                                                                                                                              |
| Abdominal TB                         | Abdominal swelling with ascites or abdominal masses<br>No acute illness                                                                                       | CXR<br>Abdominal ultrasound<br>Ascitic tap with biochemical analysis and cell counts, Xpert MTB/RIF, Xpert Ultra, or culture (if available) <sup>b</sup><br>Respiratory specimens for Xpert MTB/RIF or Ultra |
| Spinal TB                            | Deformity of spine, narrow-angle kyphosis (gibbus)<br>May have lower-limb weakness, paralysis, or loss of bowel or bladder control                            | X-ray spine<br>Fine-needle aspiration biopsy or excision biopsy with histology and Xpert MTB/RIF, Ultra or culture (if available)                                                                            |
| Pericardial TB                       | Cardiac failure<br>Distant or muffled heart sounds<br>Apex beat difficult to palpate                                                                          | CXR<br>Cardiac ultrasound<br>Pericardial tap with Xpert MTB/RIF, Xpert Ultra or culture (if available) <sup>b</sup><br>Respiratory specimens for Xpert MTB/RIF or Ultra                                      |
| TB bone and joint                    | Limitation of movement and abnormal gait<br>Unilateral joint effusion (usually knee or hip)<br>Swelling at end of long bones or small hand bones (dactylitis) | Radiograph of joint or bone<br>Joint tap or synovial biopsy, microscopy, Xpert MTB/RIF, Xpert Ultra or culture (if available) <sup>b</sup><br>Respiratory specimens for Xpert MTB/RIF or Ultra               |

# Tuberculosis of Lymph Nodes

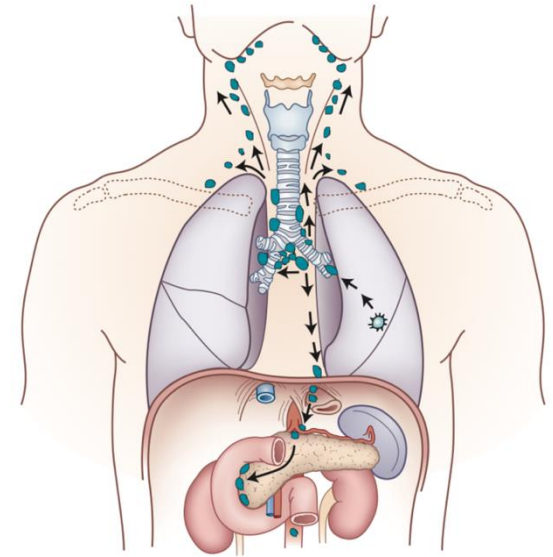
# Tuberculous Lymphadenitis

- Tuberculosis of superficial LN commonly called “Scrofula”
- The **most common extrapulmonary tuberculosis** in childhood
- Absence of specific tuberculous lymphadenitis symptoms, diagnosis is often delayed
- Can involve all regional lymph nodes, cervical lymph nodes are commonly involved
- Lymphadenitis may be the only finding in 25-35% of pediatric cases with TB



# Clinical Presentation

- Slowly progressive, painless, rubbery and discrete
- The duration of symptoms at the time of presentation is typically 1–2 months, varying from 3 weeks to 8 months
- **Cervical nodes: most common (45–70%)**, with 12–26% in the supraclavicular region
  - **Tonsillar and submandibular nodes:** most common  
-> represent extension from paratracheal LN
  - Supraclavicular node: spread from apical pleurae and upper lung field
- 20% of cases are bilateral



# Clinical Presentation

- **Non-HIV:** usually unilateral, not associated with other sites, and systemic symptoms are absent
- **HIV patients:** often associated with multifocal disease and systemic symptoms
- Concomitant pulmonary tuberculosis is reported in 18%–42% of patients with higher rates among HIV-positive patients than among HIV-negative patients
- Draining sinus may be present in 4–11% of cases



NTM



Tuberculosis

**In general clinical features do not distinguish tuberculous and NTM infection**

# Diagnosis

- A definitive diagnosis of tuberculous lymphadenitis: culture or PCR demonstration of *M. tuberculosis* in an affected lymph node
- Histologic features: nonspecific lymphoid infiltrates, noncaseating granulomas, or Langerhan giant cells in areas of extensive caseous necrosis
  - Support diagnosis of probable tuberculosis in AFB-negative, culture-negative cases

# Diagnosis

- **Fine-needle aspiration VS excisional biopsy**
  - FNA is safer, less invasive, and more practical than biopsy, especially in resource-limited settings
  - Excisional biopsy is the most invasive but has the highest sensitivity and may produce a more rapid and favorable symptomatic response
    - Rare complications of biopsy include postsurgical pain, wound infection, sinus formation, and scar

# Features of peripheral lymphadenitis due to M.Tuberculosis VS Nontuberculosis Mycobacteria

|                       | TB                                                                                                      | NTM                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| Age range (years)     | 20–40                                                                                                   | 1–6                                                       |
| Sex distribution      | F>M                                                                                                     | F≥M                                                       |
| Birth country         | TB-endemic                                                                                              | Non–TB-endemic                                            |
| HIV infection         | Common in HIV-endemic countries<br>Uncommon in developed countries                                      | Rare                                                      |
| Clinical features     | Indolent painless swelling<br>Systemic symptoms:<br>uncommon in HIV-negative,<br>common in HIV-positive | Indolent painless swelling<br>Systemic symptoms: uncommon |
| Location              | Cervical                                                                                                | Cervicofacial                                             |
| Pulmonary disease     | Common                                                                                                  | Absent                                                    |
| Tuberculin skin test  | Positive                                                                                                | Occasionally positive                                     |
| IGRA                  | Positive                                                                                                | Negative                                                  |
| Histology             | Reactive adenitis                                                                                       | Caseating granuloma                                       |
| Treatment             | Antibiotics +/- excision                                                                                | Excision +/- antibiotics                                  |
| Paradoxical reactions | Common                                                                                                  | Absent                                                    |

# Paradoxical Upgrading Reaction

- Typically, culture-negative and not associated with treatment failure
- Reported in 20%–23% of HIV-negative patients
- Occurred at a median of 1.5 months after initiation of treatment and persisted for a median of 2 months
- Manifestations of PUR: enlarging lymph nodes in 32–68%, new nodes in 27–36%, pain in 60%, and draining sinuses in 12– 60%
- These features are consistent with a robust immune response to *M. tuberculosis* with initiation of antibiotic therapy and release of mycobacterial antigens
- Steroids have been considered to reduce the robust immune response in PUR, but the use is controversial

# Management

- Therapeutic lymph node excision is not indicated except in unusual circumstances
- For large lymph nodes that are fluctuant and appear to be about to drain spontaneously, aspiration has been reported by some experts to be beneficial
- Incision and drainage have been reported to be associated with prolonged wound discharge and scarring

# Pleural Tuberculosis



# Pleural Tuberculosis



- **Second most common** form of extrapulmonary TB
- Common cause of pleural effusion in TB endemic areas
- Can be localized or generalized, typically unilateral
- Usually occur within 6-9 months of initial infection
- Most children (60-70%) have **concomitant pulmonary parenchyma disease and/or intrathoracic lymphadenopathy**
- Pleural involvement: more common in presence of HIV coinfection

# Pleural Tuberculosis



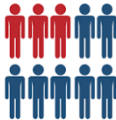
## Pathophysiology

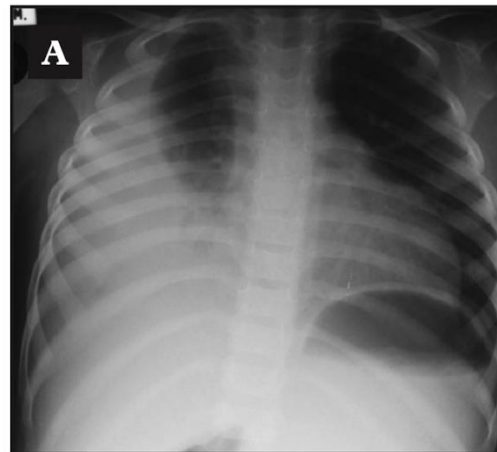
- Rupture of a subpleural tuberculous focus-> triggers an inflammatory response mediated by T- cells
- **Delayed hypersensitivity reaction** -> exudative pleural fluid
- TB bacillus invades the pleural space after the rupture of a pulmonary caseous focus (Ghon focus) in the subpleural region, by contiguity of the pulmonary lesion, by rupture of a mediastinal lymph node or via hematogenous dissemination

# Clinical Presentation

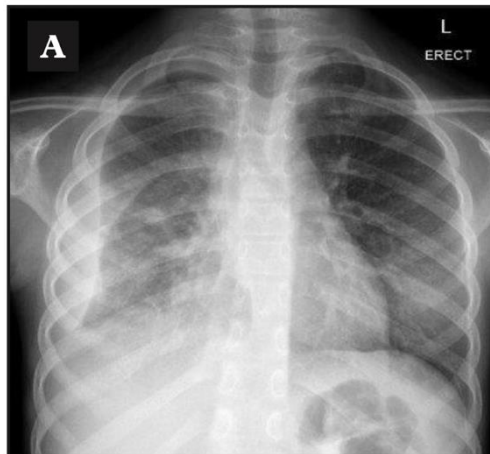


- Usually, **abrupt onset** of pleurisy resembles bacterial pneumonia
  - Fever, chest pain , shortness of breath
  - Physical examination: dullness on percussion, diminished breath sounds
- Fever is often high and persists for several weeks despite effective treatment

|                                                                                                                     |                                                                                   |               |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------|
| <b>Simple pleural effusion</b><br> |  | Fairly common |
|                                                                                                                     |  | Specific      |
|                                                                                                                     | <b>NON-SEVERE</b>                                                                 | Non-severe    |



|                                                                                                                          |                                                                                   |          |
|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------|
| <b>Complicated pleural effusion</b><br> |  | Uncommon |
|                                                                                                                          |  | Specific |
|                                                                                                                          | <b>SEVERE</b>                                                                     | Severe   |

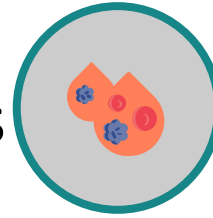


# Diagnosis



- Pleural membrane biopsy: most valuable diagnostic procedure
  - Demonstrate **caseating granulomas** in up to 90% of cases
  - Tissue culture positive 70% of cases
- Thoracentesis for pleural fluid analysis is essential
  - Exclude alternative etiologies: complicated parapneumonic bacterial pleural effusion and other infective or rarely malignant etiologies

# Pleural fluid analysis



- Fluid generally greenish yellow, occasionally tinged with blood
- Characteristically it is an **exudate with a lymphocyte predominance** and a **high protein** concentration (typically above 3 g/dl), LDH > 200 U/L, low glucose level 20-40 mg/dl
- Acid fast smear of pleural fluid is usually negative
- Mycobacterium culture positive 30-40%

# Adenosine Deaminase (ADA)

- Enzyme involved in purine metabolism
  - Typically, high concentrations in the pleural fluid in cases of pleural TB
- Pleural activity of total ADA dependent on two principal isoenzymes: ADA1, ADA2
  - ADA1 is found in all cells
  - ADA2 reflects monocyte/macrophage activation
    - Typically elevated in pleural TB
    - Values **above 40 IU/L** having high sensitivity and specificity (89 to 100%)

# Complications

- **Tuberculosis empyema**

- Chronic mycobacterial infection in pleural space
- Long standing -> thickened & calcified pleura
- Neutrophilic predominant and high burden
- Invasive surgery may be required to prevent fibrothorax



- **Pseudochylothorax:** cholesterol effusion

- **Residual pleural thickening**

- Occur in up to 50% of patients: more likely among those with initial pleural fluid analysis -> relatively low pH, low glucose concentration and high TNF-alpha concentration



# Tuberculous Meningitis



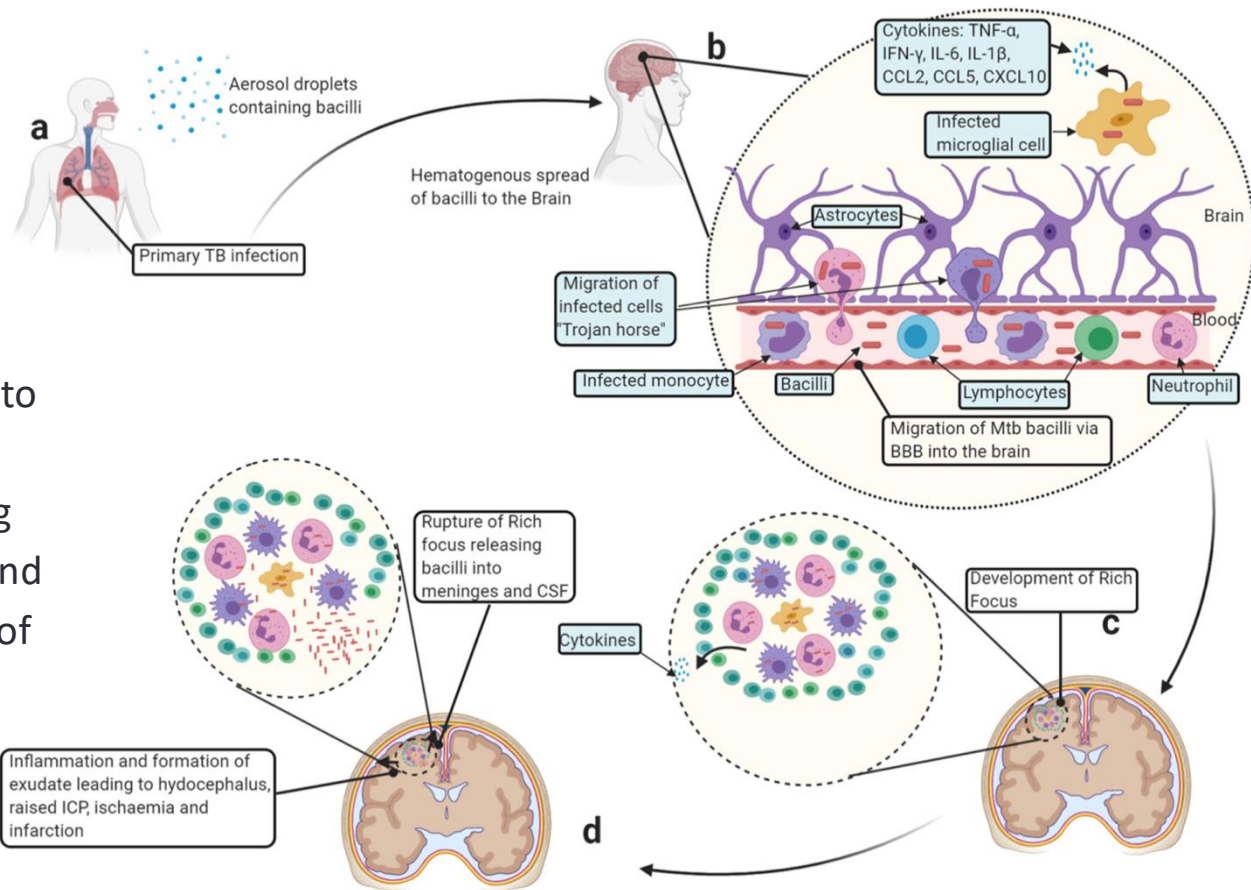
# Tuberculous Meningitis

- Common in **children** (peak incidence 2-4 years, rare in infant < 4 months of age) and **immunosuppressed** individuals such as those with HIV infection
- CNS involvement has been found in 5–10% of patients with TB and 20% of those with AIDS-related TB
- Leptomeningitis: most common CNS-TB in children and adolescents
  - Other CNS-TB diseases: intracranial tuberculoma, tuberculous brain abscesses, arachnoiditis, increased intracranial pressure, and hydrocephalus

# Pathogenesis

- Hematogenous spreading
- **“Rich foci”**
- Rupture of these lesions releases *M. tuberculosis* into the subarachnoid space or ventricular system, causing granulomatous infection and subsequent inflammation of meninges

On an average, TBM occurs 6 to 12 months after the primary infection





## Pathogenesis

In TBM, a thick gelatinous exudate develops around the basal cisterns, the sylvian fissure and the brainstem

As this may cause obstruction to the outflow from the fourth ventricle or indeed from the cerebral aqueduct, **hydrocephalus** frequently develops in these children,

Further exacerbates the clinical condition by raising the **intracranial pressure (ICP)**



# Presentation

- Clinical onset may be acute, subacute or gradual and is characterized by non-specific symptoms in the early stages
  - Low-grade fever, malaise, headache, dizziness, vomiting, personality changes, and symptoms related to pulmonary TB (such as cough)
- Advanced disease:
  - Signs of meningeal irritation, raised ICP (bulging fontanelle, sunset sign, papilledema), cranial nerve palsies, neurological deficits, altered sensorium and movement disorders



# Presentation

- British Medical Research Council (BMRC) staging of the disease
  - Stage I GCS 15 without focal neurological deficits
    - Nonspecific symptoms: fever, headache, irritability
    - Typically last 1-2 weeks
  - Stage II GCS 11-14 with focal neurological deficits
    - Abruptly lethargy and apathy, nuchal rigidity, hypertonia, vomiting, cranial nerves abnormalities
  - Stage III GCS  $\leq 10$  stupor, seizures, coma, hemiplegia or paraplegia
    - Decerebrate or decorticate posturing, progressive vital sign abnormalities and eventually death



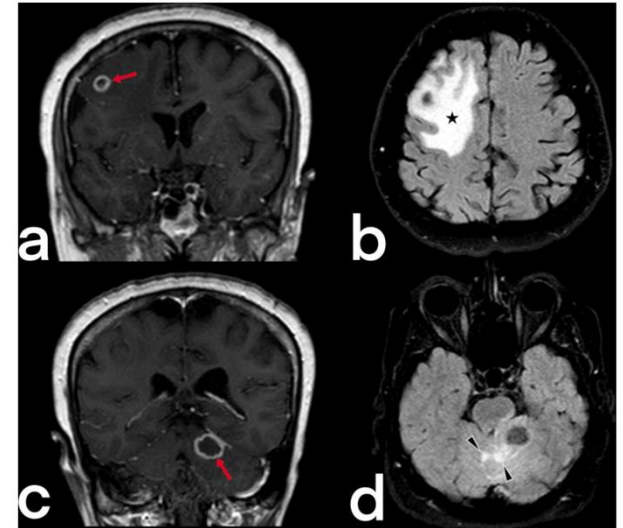
# Neurocomplication

- Progressive hydrocephalus, vasculitis, infarction, and cranial neuropathies
- **Most common: communicating hydrocephalus**, which is caused by the obstruction of cerebrospinal fluid into the basal cisternae
- Some cases, non-communicating hydrocephalus caused by a tuberculoma or tuberculous abscess may also be seen
- Ischemic infarction may also develop as a result of cerebral arteritis
- Hemorrhagic infarction caused by dural venous sinus thrombosis
- Cranial nerves 2, 3, 4, and 7 may be affected by vascular ischemia or nerve entrapment



# Tuberculoma

- Firm, avascular, spherical masses, with size varying between 2 cm and 10 cm in diameter
  - Center contains necrotic caseous material surrounded by a capsule comprising fibroblasts, epithelioid cells, Langhans giant cells, and lymphocytes
- Lesions: solitary, multiple, or miliary
- **Frontal and parietal** lobes are often involved
- Imaging typical finding: central nidus of calcification with surrounding ring-like enhancement, known as the **target sign**







# Diagnosis

- **CSF analysis**

- Leukocytosis with lymphocyte predominance, protein elevation, decrease in CSF glucose (hypoglycorrhachia)

- **Bacteriology**

- CSF smears for AFB has poor sensitivity (10-15%)
- Mycobacterial culture: gold standard, higher sensitivity (50% - 60%) but limitation in turnaround time (up to 8 weeks)
- XpertMTB/RIF: specificity 94% and sensitivity 54 % in children

**WHO recommendation:** Children with signs and symptoms of TB meningitis, **Xpert MTB/RIF or Xpert Ultra should be used in CSF as an initial diagnostic test for TB meningitis** rather than smear microscopy/culture



# Diagnosis

- **Neuroimaging**

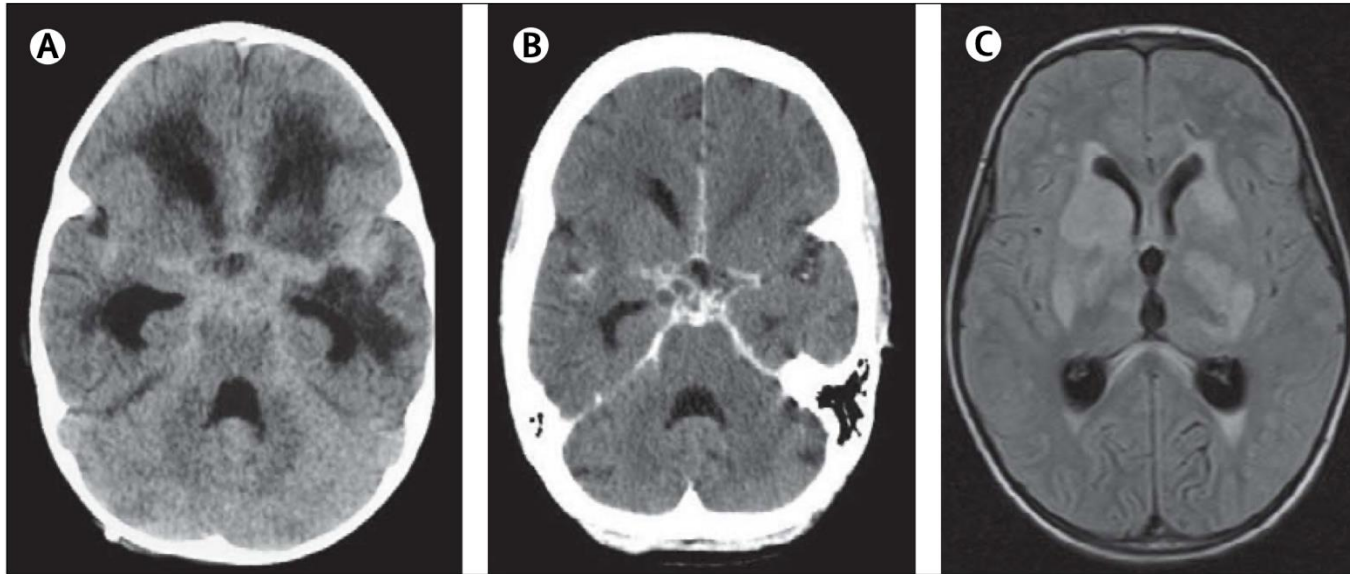
- CT brain

- **Hydrocephalus and basal meningeal enhancement** are the most common radiological features of tuberculous meningitis
    - Basal meningeal enhancement was the most sensitive feature (89%)

- MRI brain

- Higher sensitivity than CT for the detection of abnormalities such as meningeal enhancement, infarcts, and tuberculomas

## Cerebral imaging abnormalities observed in tuberculous meningitis



(A). Non-contrast CT scan: hyperdense material occupying the suprasellar and middle cerebral artery cisterns and hydrocephalus with periventricular lucency

(B). Post-contrast CT scan showing basal meningeal enhancement and multiple right-sided suprasellar rim-enhancing lesions

(C). MRI flair sequence showing bilateral basal ganglia infarcts

# Outcome

- The risk of **death** among children aged 0–14 years with TBM was estimated at **19.3%**
- The risk of **neurological sequelae** among TBM survivors was estimated at **36.7%**

# Skeletal Tuberculosis



# Skeletal Tuberculosis

- Bone and joint tuberculosis occur in 1-5% of children with tuberculosis
  - Spine is involved in up to 50% of skeletal TB cases: tuberculous spondylitis , femur, long bone around knee and ankle
  - Less frequently: ribs, mandible, sternum, clavicle
  - Multifocal osteomyelitis reported in 10-15% of cases
- Causes bone destruction, spinal deformity and neural complications
- Disease becomes symptomatic during first 1-3 years of initial infection
  - Each most frequently involved bone and joints has characteristic incubation period: 1 month for dactylitis, 30 months for tuberculosis of hip



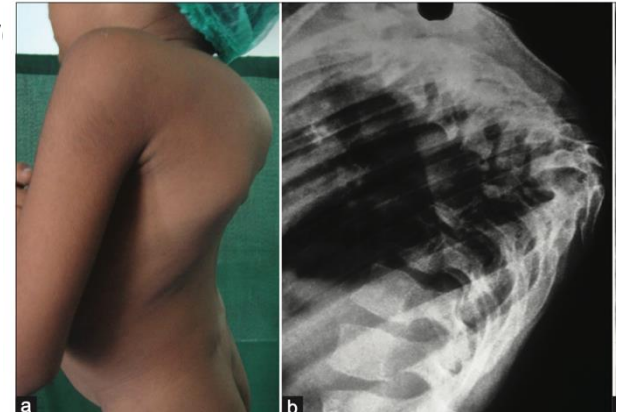
# Pathophysiology

- Hematogenous or lymphatic dissemination
  - Skeletal tuberculosis occur more in young children than adult due to high blood flow through growing bone
  - Lesion usually start in area of endarteritis in metaphysis of long bone (abundant blood supply)
- Bone infection can be initiated by other two mechanism especially spine
  - Direct extension through lymphatic from caseous paravertebral LN
  - Direct local hematogenous or lymphatic extension from neighboring bone
- Bone is destroyed progressively by pressure necrosis and formation of cold abscess



## Clinical Features

- Insidious onset, slow course and mild initial symptoms
- Low grade fever, weight loss, pain that worse at night and soft tissue swelling at site of infection
- Vertebral involvement begins in the **anterior vertebral body**, eventually causing disk space collapse and **anterior wedging** of vertebral bodies sometimes causing “**gibbus deformity**” (angulation of spine), extensive severe kyphosis (**Pott disease**)
- Usual site: **lower thoracic spine**, lumbar spine
  - Usually two vertebrae are involved (contiguous>skip lesion)







## Complications

- Paravertebral abscess (Pott abscess)
- Retropharyngeal abscess
- Psoas abscess
- Neurological complication 10-30%: caused by inflammation of the spinal cord secondary to a neighboring cold abscess, by caseum or granuloma in extradural space, or by spinal vessel thrombosis
  - Most often arise from cervical and lumbar vertebral lesions
  - Neuroplegia, paraplegia or even quadriplegia



# Investigation

- TST/IGRA are usually positive
- Plain film: periarticular osteopenia, lytic lesion in the body of the vertebra , joint space narrowing and soft tissue swelling
- Chest X-ray often normal
- **CT scan:** useful for evaluation bone destruction, adjacent soft tissue abscess formation and calcification, and in guiding percutaneous biopsy
- MRI helpful in determining extent of bone and soft tissue disease
- Biopsy specimens should be obtained to demonstrate organism and culture

In comparison to pyogenic discitis, the most distinguishing feature of spinal TB is bony destruction with relative preservation of the intervertebral disc and heterogeneous enhancement

# Management

- Uncomplicated cases are managed with medical rather than surgical treatment
- Surgery can be considered in situations in which
  - (1) Poor response to chemotherapy with evidence of ongoing infection or ongoing deterioration
  - (2) Relief of cord compression is needed in patients with persistence or recurrence of neurologic deficits
  - (3) Instability of the spine



Treatment

# Treatment Regimens for Extrapulmonary TB

| Age and type of EPTB                                       | Treatment regimen <sup>a</sup> |                    |
|------------------------------------------------------------|--------------------------------|--------------------|
|                                                            | Intensive phase                | Continuation phase |
| <b>Infants aged &lt;3 months or weighing &lt;3 kg</b>      |                                |                    |
| Peripheral lymph node TB                                   | 2HRZ or 2HRZE <sup>b</sup>     | 4HR                |
| <b>Children and adolescents aged 3 months–&lt;16 years</b> |                                |                    |
| Peripheral lymph node TB                                   | 2HRZ or 2HRZE <sup>b</sup>     | 2HR                |
| <b>Adolescents aged &gt;16 years</b>                       |                                |                    |
| Peripheral lymph node TB                                   | 2HRZ or 2HRZE <sup>b</sup>     | 4HR                |
| <b>Children and adolescents aged 0–19 years</b>            |                                |                    |
| EPTB <sup>c</sup>                                          | 2HRZE                          | 4HR                |
| TBM <sup>d</sup> (strong recommendation)                   | 2HRZE                          | 10HR               |
| TBM <sup>d</sup> (conditional recommendation)              | 6HRZEto                        |                    |
| Osteoarticular TB                                          | 2HRZE                          | 10HR               |

# Treatment for Drug-susceptible TB

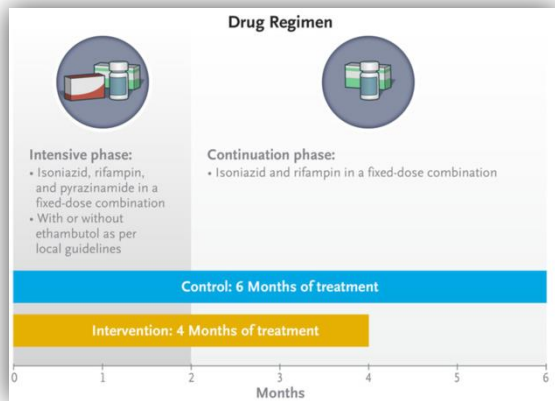


## Shorter Treatment for Nonsevere Tuberculosis in African and Indian Children

Turkova A et al. DOI: 10.1056/NEJMoa2104535

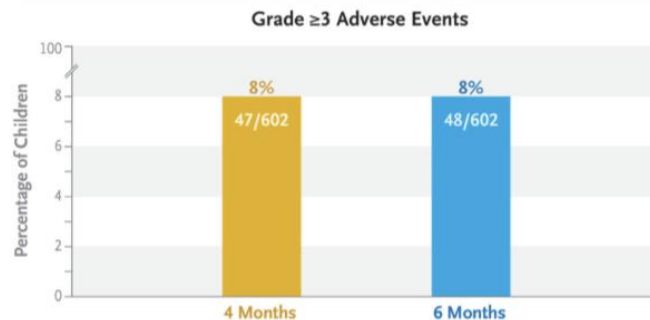
- Children age **3 months -16 years**
- Symptomatic **non-severe** tuberculosis, **smear negative** on a respiratory sample

2HRZE/4HR VS 2HRZ(E)/2HR



### Non-severe tuberculosis

- respiratory tuberculosis confined to one lobe (opacification of <1 lobe)
- no cavities
- no signs of miliary tuberculosis
- no complex pleural effusion
- no clinically significant airway obstruction
- peripheral lymph-node tuberculosis



### CONCLUSIONS

Among children with nonsevere, drug-susceptible, smear-negative TB, a 4-month treatment regimen was noninferior to a 6-month regimen at 72 weeks of follow-up.

## Recommendation 7.

In children and adolescents between 3 months and 16 years of age with non-severe TB (without suspicion or evidence of MDR/RR-TB), a 4-month treatment regimen (2HRZ(E)/2HR) should be used (strong recommendation, moderate certainty of evidence) – new recommendation.

### Non-severe TB

- **Peripheral lymph node TB**
- Intrathoracic lymph node TB without airway obstruction
- **Uncomplicated TB pleural effusion** (without pneumothorax or empyema)
- Paucibacillary
- Non-cavitary disease, confined to one lobe of the lungs without a miliary pattern

# Recommended dosages of First-line TB medicines for use in children and young adolescents

| Medicine         | Dose (mg/kg body weight) | Range (mg/kg body weight) |
|------------------|--------------------------|---------------------------|
| Isoniazid (H)    | 10                       | 7–15 <sup>a</sup>         |
| Rifampicin (R)   | 15                       | 10–20                     |
| Pyrazinamide (Z) | 35                       | 30–40                     |
| Ethambutol (E)   | 20                       | 15–25                     |



# Treatment for TB Meningitis

## WHO Recommendation

In children and adolescents with bacteriologically confirmed or clinically diagnosed TB meningitis (without suspicion or evidence of MDR/RR-TB)

**12-month regimen (2HRZE/10HR)** Recommended doses to be used in this regimen are the same as those for the treatment of PTB

**6-month intensive regimen (6HRZeto)** may be used as an alternative option

## Recommend Dosing for **6HRZEto**

| Medicine          | Recommended dose range in the interim dosing strategy<br>(mg/kg body weight) |
|-------------------|------------------------------------------------------------------------------|
| Isoniazid (H)     | 15–20 <sup>a</sup>                                                           |
| Rifampicin (R)    | 22.5–30                                                                      |
| Pyrazinamide (Z)  | 35–45                                                                        |
| Ethionamide (Eto) | 17.5–22.5                                                                    |

- **Higher dose of INH and rifampicin**
- **Greater CNS penetration of Eto than ETB**

# Indications for Adjuvant Therapy

## Corticosteroids

- No WHO recommendations regarding use of corticosteroids for forms of EPTB disease other than **TBM and tuberculous pericarditis**
- Sometimes used for complicated forms of TB (e.g. complications of airway obstruction by TB lymph nodes; severely ill children and adolescents with disseminated TB)
  - Prednisone dosage of 2 mg/kg/day orally, increased to 4 mg/kg/day in severely ill children and adolescents (e.g. with danger signs), with a maximum dosage of 60 mg/day for 4 weeks
  - Reduced gradually over 2–4 weeks before stopping.
  - Alternatively, dexamethasone 0.3–0.6 mg/kg/day can be used with a similar duration and by using the same method of gradual reduction in the dose

# Adjuvant Steroids in the Treatment of TB Meningitis and pericarditis

## Recommendation

In patients with tuberculous meningitis, an initial adjuvant corticosteroid therapy with dexamethasone or prednisolone tapered over 6–8 weeks should be used (strong recommendation, moderate certainty of evidence).

## Recommendation

In patients with tuberculous pericarditis, an initial adjuvant corticosteroid therapy may be used (conditional recommendation, very low certainty of evidence).

# Thank You

**END TB**

**NO STIGMA**

**END TB**

